

Question	6	Benchmark(s)	MA.912.AR.3.1				
<table><tr><td>Work</td><td>$12(x - 6)^2 + 3922 = 5650$$12(x - 6)^2 = 1728$$(x - 6)^2 = 144$$x - 6 = \pm 12$$x - 6 = 12 \quad x - 6 = -12$$x = 18 \quad x = -6$</td></tr><tr><td>Year(s)</td><td>2028</td></tr></table>		Work	$12(x - 6)^2 + 3922 = 5650$ $12(x - 6)^2 = 1728$ $(x - 6)^2 = 144$ $x - 6 = \pm 12$ $x - 6 = 12 \quad x - 6 = -12$ $x = 18 \quad x = -6$	Year(s)	2028	Recommended Scoring (18 Total Points)	Consider giving 8 points for the correct year. Consider giving 4 of those 8 points for a student who includes the year 2004 or gives the answer as 18. A student can state that the answer is 18 years since 2010. Consider distributing the other 10 points for correct work.
		Work	$12(x - 6)^2 + 3922 = 5650$ $12(x - 6)^2 = 1728$ $(x - 6)^2 = 144$ $x - 6 = \pm 12$ $x - 6 = 12 \quad x - 6 = -12$ $x = 18 \quad x = -6$				
		Year(s)	2028				
Notes	It is possible for students to choose to solve by factoring or using the quadratic formula. Carefully examine student work to consider partial work.						